

TASK 1 REPORT

Foundation & Environment Setup – Cybersecurity Internship

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Task: Foundation & Environment Setup

Introduction

The objective of this task is to build strong fundamentals in cybersecurity, networking, cryptography, and set up a professional cybersecurity lab environment. During this task, I learned cybersecurity concepts, Linux basics, networking fundamentals, cryptography concepts, and familiarized myself with important cybersecurity tools.

1.Cybersecurity Basics

1. CIA Triad

Confidentiality

Confidentiality ensures that information is accessible only to authorized individuals. It protects sensitive data from unauthorized access.

Example: Password-protected accounts and encrypted files.

Integrity

Integrity ensures that data remains accurate, complete, and unchanged unless modified by authorized users.

Example: File hashing and digital signatures.

Availability

Availability ensures that systems, applications, and data are accessible whenever needed by authorized users.

Example: Backup servers and disaster recovery systems.

2. Threat Types

Phishing

A cyberattack where attackers use fake emails, websites, or messages to steal sensitive information such as passwords and banking details.

Malware

Malicious software designed to damage, disrupt, or gain unauthorized access to computer systems.

DDoS (Distributed Denial of Service)

An attack that overwhelms a server or network with excessive traffic, making services unavailable to legitimate users.

SQL Injection

A web attack that inserts malicious SQL commands into application inputs to access or manipulate databases.

Brute Force Attack

A method of gaining access by repeatedly trying different password combinations until the correct one is found.

Ransomware

A type of malware that encrypts files or locks systems and demands payment to restore access.

3. Attack Vectors

Social Engineering

Manipulating people into revealing confidential information or performing actions that compromise security.

Wireless Attacks

Attacks targeting wireless networks, including unauthorized access to Wi-Fi networks and interception of wireless communications.

Insider Threats

Security risks originating from employees, contractors, or other trusted individuals who misuse their access privileges.

Conclusion

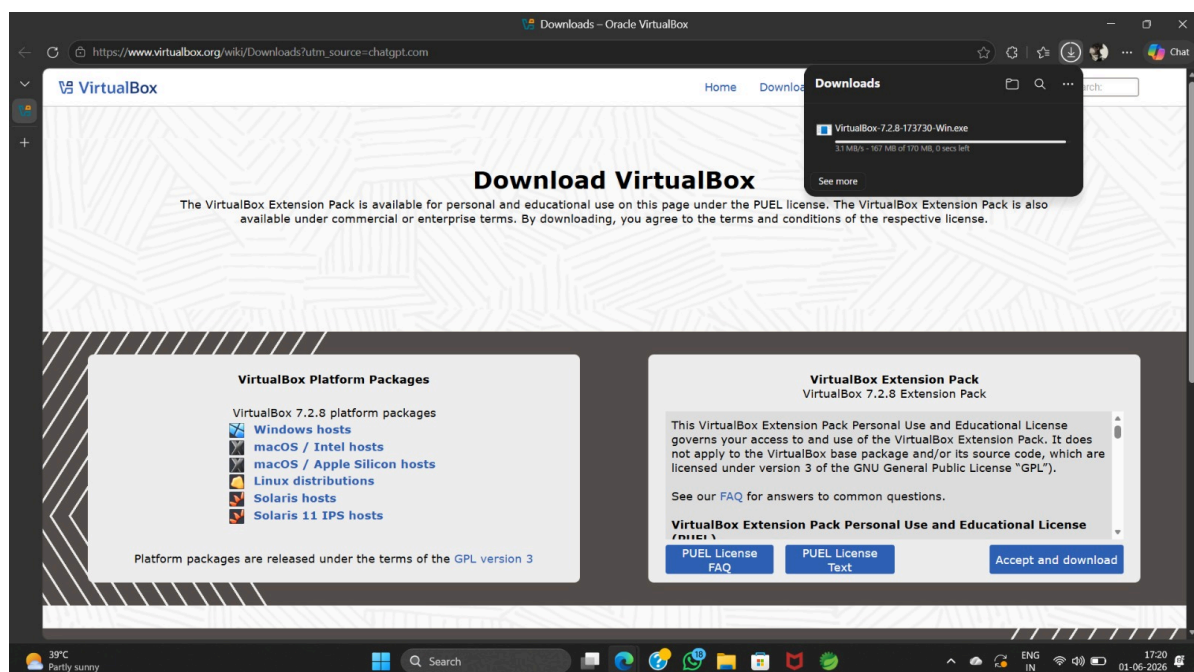
Cybersecurity focuses on protecting systems, networks, and data from digital attacks. Understanding the CIA Triad, common threats, and attack vectors forms the foundation of cybersecurity knowledge.

Document Link : [Cybersecurity Basics - Google Docs](#)

2. Lab Environment Setup

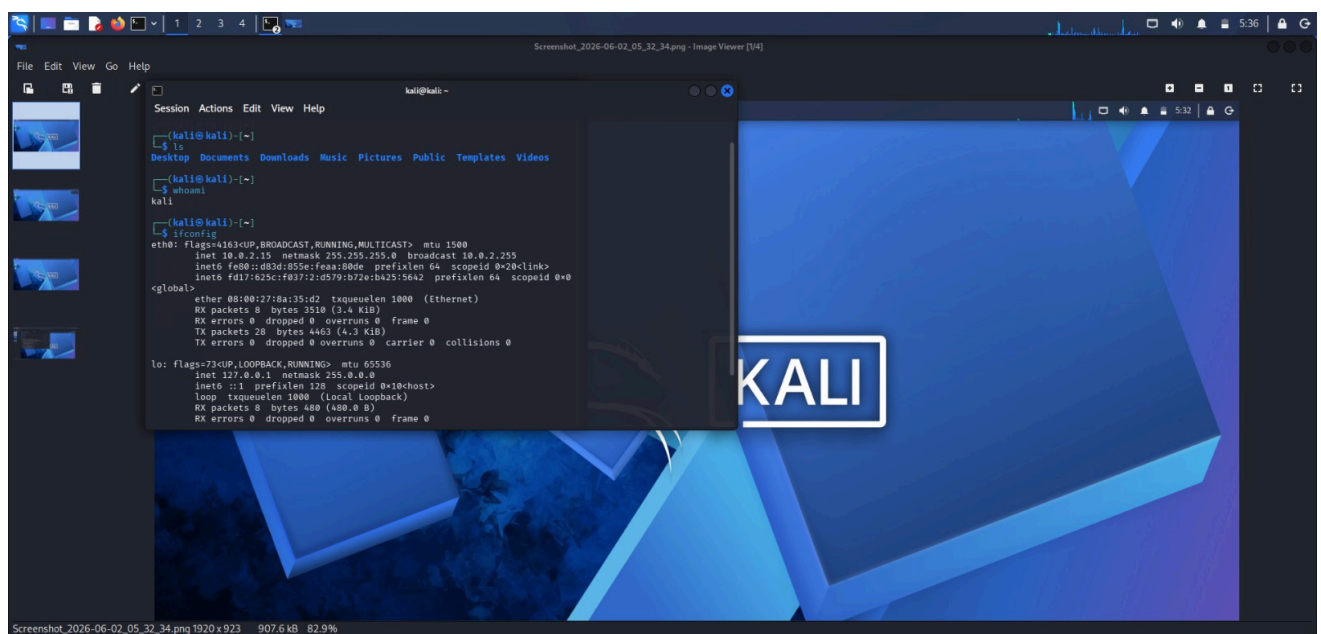
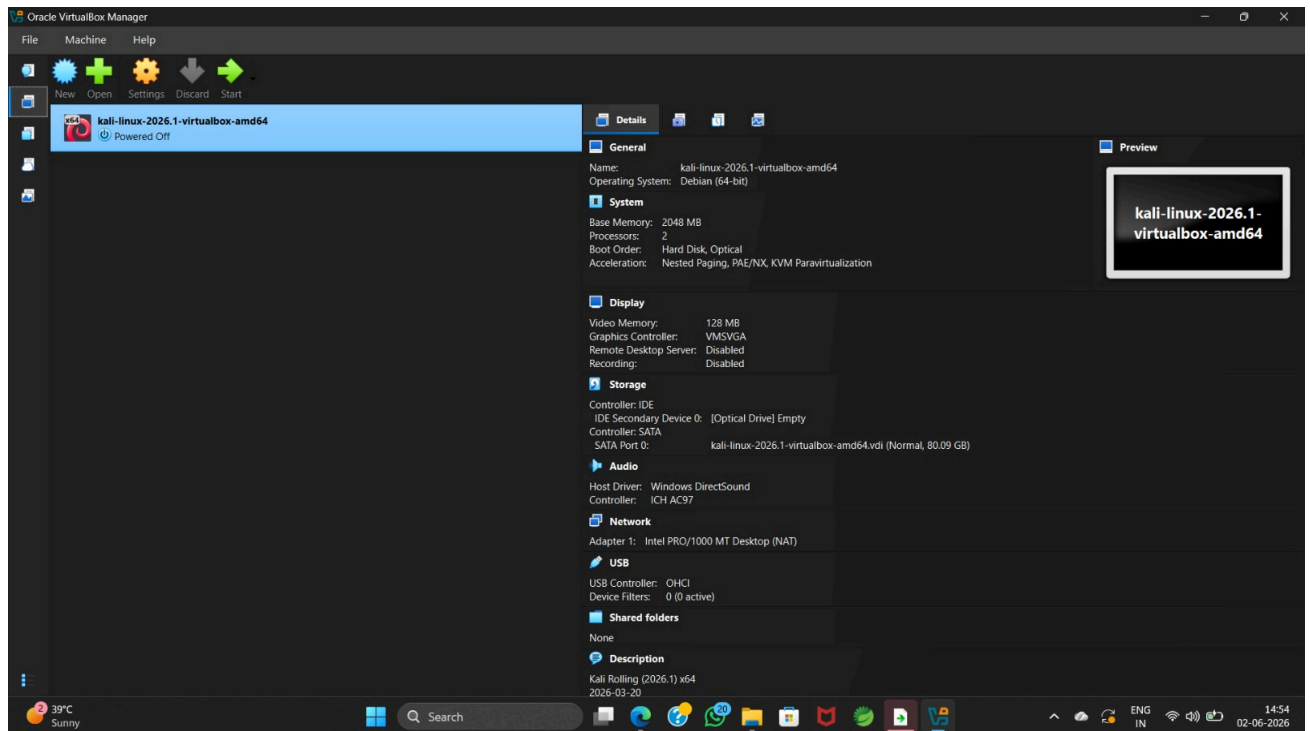
2.1 VirtualBox Installation

Oracle VirtualBox was installed to create virtual machines for cybersecurity testing and learning.



3.2 Kali Linux Setup

Kali Linux was installed and configured as the attacker machine.



3.3 Host-Only Adapter Configuration

A Host-Only Adapter was configured to create an isolated lab environment for secure testing.

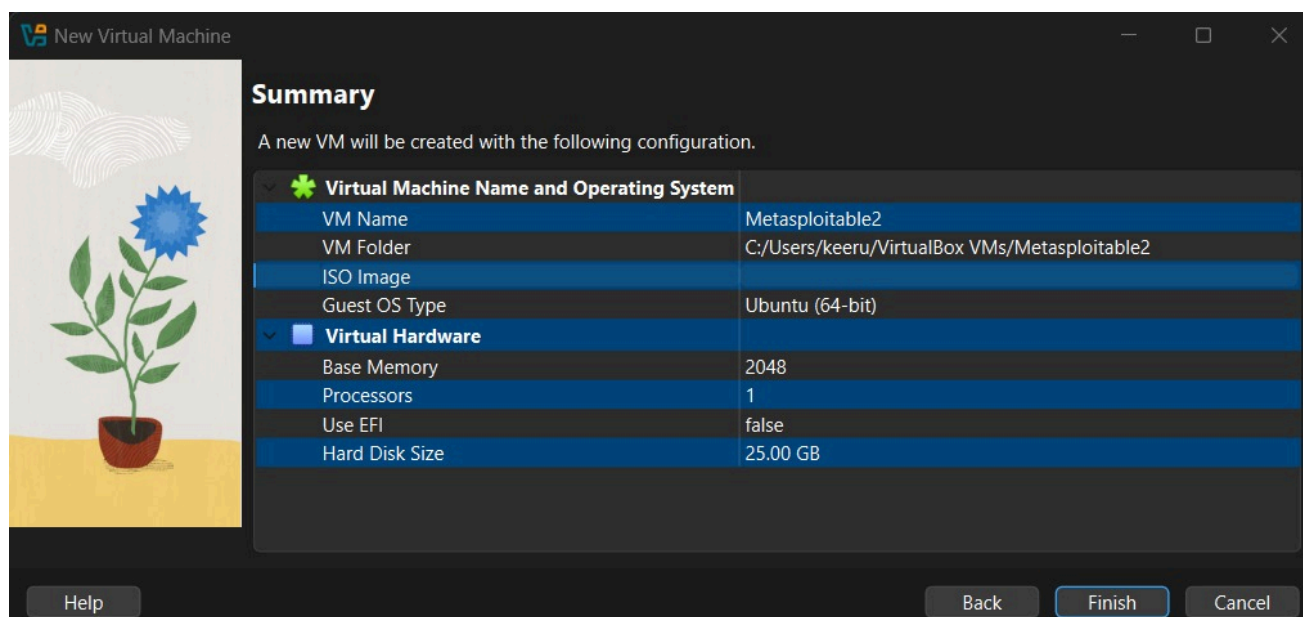
```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox : 1
File Machine View Input Devices Help

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$ ping 192.168.112.3
PING 192.168.112.3 (192.168.112.3) 56(84) bytes of data.
64 bytes from 192.168.112.3: icmp_seq=1 ttl=64 time=3.36 ms
64 bytes from 192.168.112.3: icmp_seq=2 ttl=64 time=2.38 ms
64 bytes from 192.168.112.3: icmp_seq=3 ttl=64 time=1.55 ms
64 bytes from 192.168.112.3: icmp_seq=4 ttl=64 time=3.39 ms
64 bytes from 192.168.112.3: icmp_seq=5 ttl=64 time=1.63 ms
64 bytes from 192.168.112.3: icmp_seq=6 ttl=64 time=5.63 ms
64 bytes from 192.168.112.3: icmp_seq=7 ttl=64 time=2.54 ms
64 bytes from 192.168.112.3: icmp_seq=8 ttl=64 time=3.30 ms
64 bytes from 192.168.112.3: icmp_seq=9 ttl=64 time=1.16 ms
64 bytes from 192.168.112.3: icmp_seq=10 ttl=64 time=1.26 ms
64 bytes from 192.168.112.3: icmp_seq=11 ttl=64 time=1.46 ms
64 bytes from 192.168.112.3: icmp_seq=12 ttl=64 time=1.75 ms
64 bytes from 192.168.112.3: icmp_seq=13 ttl=64 time=1.66 ms
64 bytes from 192.168.112.3: icmp_seq=14 ttl=64 time=2.38 ms
^C
— 192.168.112.3 ping statistics —
14 packets transmitted, 14 received, 0% packet loss, time 13036ms
rtt min/avg/max/mdev = 1.156/2.387/5.625/1.169 ms
```

3.4 Metasploitable2 / DVWA Setup

A vulnerable target environment was prepared for cybersecurity practice.



4. Linux Fundamentals

Linux is one of the most important operating systems used in cybersecurity.

Commands Studied

pwd

Displays the current working directory.

ls

Lists files and folders.

cd

Changes the current directory.

chmod

Changes file permissions.

apt

Used for package management.

ifconfig

Displays network configuration information.

For More Info:- Document Link with their working screenshots :[Linux Fundamentals - Google Docs](#)

5. Networking Basics

5.1 OSI Modelin

The OSI Model consists of seven layers:

1. Physical Layer
2. Data Link Layer

3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

5.2 TCP/IP Protocol Suite

The TCP/IP model contains:

- Application Layer
- Transport Layer
- Internet Layer
- Network Access Layer

5.3 DNS

DNS (Domain Name System) converts domain names into IP addresses.

Example:

google.com → IP Address

5.4 HTTP and HTTPS

HTTP is used for web communication.

HTTPS is the secure version of HTTP and uses SSL/TLS encryption.

5.5 IP Addressing, Subnetting, and NAT

IP Addressing

Used to uniquely identify devices on a network.

Subnetting

Divides a network into smaller logical networks.

NAT

Network Address Translation converts private IP addresses into public IP addresses.

Document Link(My Notes) : [Networking Basics - Google Docs](#)

6. Cryptography Basics

Cryptography is used to protect information and secure communications.

6.1 Symmetric Encryption

Symmetric encryption uses the same key for both encryption and decryption.

Advantages:

- Fast
- Efficient for large amounts of data

Disadvantages:

- Key sharing can be difficult

Examples:

- AES
- DES

6.2 Asymmetric Encryption

Asymmetric encryption uses two keys:

- Public Key
- Private Key

Advantages:

- More secure key exchange
- Supports digital signatures

Disadvantages:

- Slower than symmetric encryption

6.3 Hashing

Hashing is used to verify data integrity.

Hashing converts data into a fixed-length value called a hash.

Characteristics:

- One-way process
- Cannot be reversed
- Used for password storage and integrity verification

Examples:

- MD5
- SHA256

6.4 Digital Certificates and SSL/TLS

Digital certificates verify the identity of websites and organizations.

Functions:

- Authentication
- Encryption support
- Trust establishment

SSL (Secure Sockets Layer) and TLS (Transport Layer Security) secure communication over networks.

Benefits:

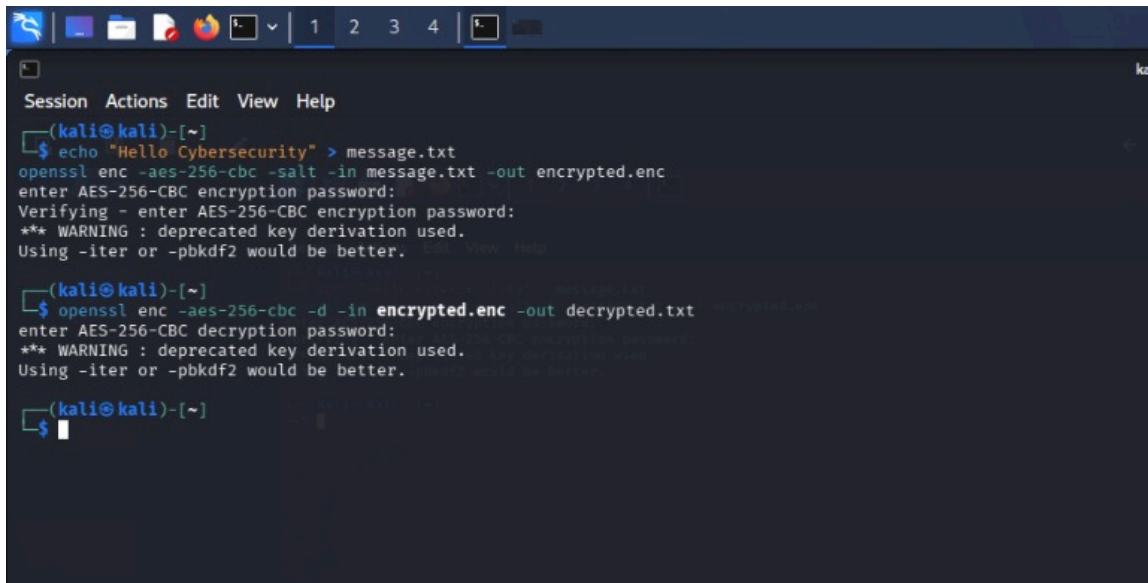
- Encrypts transmitted data
- Protects sensitive information
- Used in HTTPS websites

6.5 OpenSSL Demonstration

OpenSSL is an open-source toolkit used for encryption and security operations.

Common Uses:

- Generate certificates
- Encrypt files
- Decrypt files
- Create hashes



```
Session Actions Edit View Help
(kali@kali)-[~]
└─$ echo "Hello Cybersecurity" > message.txt
openssl enc -aes-256-cbc -salt -in message.txt -out encrypted.enc
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(kali@kali)-[~]
└─$ openssl enc -aes-256-cbc -d -in encrypted.enc -out decrypted.txt
enter AES-256-CBC decryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(kali@kali)-[~]
└─$
```

Conclusion

Cryptography is essential for protecting data, securing communications, and ensuring privacy in modern computer systems.

7. Tool Familiarization

7.1 Wireshark

Purpose: Packet capture and network traffic analysis.

7.2 Nmap

Purpose: Network scanning and host discovery.

Command Used: nmap localhost

7.3 Burp Suite

Purpose: Web application security testing and web proxy analysis.

7.4 Netcat

Purpose: Network debugging and troubleshooting.

Command Used: nc -h

```

kali@kali:~$ sudo apt autoremove
Session Actions Edit View Help

kali@kali:~$ sudo apt autoremove
** (Wireshark:12304) 03:18:40.207815 (Capture MESSAGE) -- Capture Start ...
** (Wireshark:12304) 03:18:40.279928 (Capture MESSAGE) -- Capture started
** (Wireshark:12304) 03:18:40.280351 (Capture MESSAGE) -- File: /tmp/Wireshark_eth0#P8BQJ.pcapng*
** (Wireshark:12304) 03:18:40.430775 (Capture MESSAGE) -- Capture Stop ...
** (Wireshark:12304) 03:18:54.062512 (Capture MESSAGE) -- Capture stopped.

kali@kali:~$ sudo apt install nmap -y
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
binutils-mingw-w64-base mingw-w64-1008-dev python3-pefile
binutils-mingw-w64-1008 mingw-w64-x86-64-dev python3-poetry-core
binutils-mingw-w64-x86-64 python3-requests python3-poetry-dynamic-versioning
bloodhound.py python3-ardour python3-pyexploitdb
comerr-dev python3-aesdub python3-pyfiglet
dmspp python3-aiocore python3-pylint
dniff python3-aiocompose python3-pynfsclient
ettercap-common python3-aiosmb python3-pyrsr
ettercap-graphical python3-aiowinreg python3-pyysatz
figlet python3-irc4 python3-psychodan
finger python3-ecscitree python3-pysnmp
gcc-mingw-w64-base python3-asmitools python3-pyservio
gcc-mingw-w64-1008-win32 python3-asynauth python3-qasync
gcc-mingw-w64-1008-win32-runtime python3-asynctools python3-qrcode
gcc-mingw-w64-x86-64-win32 python3-bitstruct python3-serial-asyncio
gcc-mingw-w64-x86-64-win32-runtime python3-browser-cookie3 python3-tmap
imagemagick python3-dploit python3-tls
imagemagick-7_q16 python3-dunamai python3-tomlkit
kbs-m2-dev python3-git python3-unikrypto
libapache2-mod-php python3-gitdb python3-wapiti-swagger
libgsppc4t64 python3-httpx-ntln python3-winact
libwasmcint-nt12 python3-jeepey python3-wmtdict
libkademl5rv-ml12 python3-lussy python3-rsh-redone-client
libkademl5rv python3-masy sntp-user-enum
libkps-dev python3-minisump scripta-scripts
liblajit-5.1-2 python3-minikerberos toilet-fonts
liblajit-5.1-1-common python3-mldap unicorncan
libnls1-21t64 python3-neo4j urlican
libnls2-6 python3-neobit wapiti
libnls3 python3-neoline
mingw-w64-common python3-oscrypto

Use 'sudo apt autoremove' to remove them.

Upgrading:
editf nmap-common

Installing:
nmap

```

Kali Linux terminal session showing a successful nmap scan of localhost and subsequent actions.

```

kali@kali:~$ nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-06 03:20 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000038s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds

kali@kali:~$ nc -h
nc -h
[11:50:11]
connect to somewhere: nc [-options] hostname port[s] [ports] ...
listen for inbound: nc -l -p port [-options] [hostname] [port]
options:
  -c shell commands          as "-e"; use /bin/sh to exec [dangerous!!]
  -f filename                allow broadcasts
  -g gateway                 source-routing hop point[s], up to 8
  -h num                     source-routing pointer: 4, 8, 12, ...
  -i secs                   this cruft
  -l                          delay interval for lines sent, ports scanned
  -L                          get keepalive option on socket
  -l                          listen mode, for inbound connects
  -n                          numeric-only IP addresses, no DNS
  -n                          hex dump of traffic
  -p port                    local port number
  -P                          randomly local and remote ports
  -q secs                    quit after EOF on stdin and delay of secs
  -a addr                    local source address
  -t tos                      set Type of Service
  -u                          answer TELNET negotiation
  -v                          NOP mode
  -V                          verbose [use twice to be more verbose]
  -w secs                    timeout for connects and final net reads
  -C                          Send CRLF as line-ending
  -Z                          zero-I/O mode [used for scanning]
port numbers can be individual or ranges: lo-hi [inclusive];
hyphen in port names must be backslash escaped (e.g. "ftp-data").

kali@kali:~$ burpsuite
[warning] user/bin/burpsuite: No JAVA_CMD set for run_java, falling back to JAVA_CMD = java
SLF4J(W): No SLF4J providers were found.
SLF4J(W): Defaulting to no-operation (NOP) logger implementation
SLF4J(W): See https://www.slf4j.org/codes.html#noProviders for further details.
SLF4J: No SLF4J providers were found.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See https://www.slf4j.org/codes.html#noProviders for further details.
  
```

Document Link: [TOOL FAMILIARIZATION - Google Docs](#)

8. Conclusion

In this task, I learned the fundamentals of cybersecurity, networking, cryptography, and Linux. I successfully set up a cybersecurity lab environment using VirtualBox and Kali Linux. I also explored networking concepts, encryption techniques, and cybersecurity tools including Wireshark, Nmap, Burp Suite, and Netcat. This task

helped me build a strong foundation for future cybersecurity learning and practical activities.

References

- [Kali Linux Documentation](#)
- [Wireshark Documentation](#)
- [Nmap Documentation](#)
- [Burp Suite Documentation](#)
- [OpenSSL Documentation](#)
- [TCP/IP and Networking Fundamentals](#)